

Rates and Unit Rates

Lesson 4-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

\$12 for 4 games is a rate

Rate

A ratio comparing two amounts with different units, like miles per hour.

\$3 per 1 game

Unit Rate

A rate for just 1 of something, like cost for 1 item.

60 miles per hour means 60 miles for every 1 hour

Per

For each one. Example: 5 dollars per book.

5 to 3 or 5:3

Ratio

A way to compare two amounts.

Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil — Pack B is the better buy

Better Buy

The choice that costs less for each item.

$\$12 \div 4 \text{ games} = \3 per game

Divide

Splitting a total into equal groups.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find a unit rate to compare prices and decide the better buy.

1 Notice

You're comparing ticket prices at different game booths in the arcade.

2 Key idea

I can find a unit rate to compare prices and decide the better buy.

3 Words I'll use

Rate

Unit Rate

Per

Ratio

Better Buy

Divide



Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Booth A charges \$3 for 5 games and Booth B charges \$5 for 8 games. Just by looking, which booth seems like the better deal, and what makes it hard to be sure?

Level 1 support

Start here: It is hard to compare because the booths have different numbers of games.

 **Need a hint?**

Sentence starters (optional)

Booth ____ looks better because ____.

El puesto ____ se ve mejor porque ____.

It is hard to compare because the ____ are different.

Es difícil comparar porque los/las ____ son diferentes.

Word bank: rate per ratio compare

Listen for: Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

Level 2

Why can you not just pick the booth with the lower total price of \$3? Explain what a fair comparison actually needs.

- You cannot just use the lower price because ____.
- A fair comparison needs ____ so that ____.

EXPLORE

How did you find the cost per game (the unit rate) for each booth, and what did you divide?

Level 1 support

Start here: I divided the total cost by the number of games to get the cost for one game.



Need a hint?

Sentence starters (optional)

The unit rate is ____ per one ____.

La tasa unitaria es ____ por un ____.

I divided ____ by ____ to get the cost per game.

Dividí ____ entre ____ para obtener el costo por juego.

Word bank: **unit rate** **divide** **per** **rate** **cost**

Listen for: Students explain they divided total cost by number of games (e.g., $\$5.00 \div 8 = \0.625) and state the result as a cost per one game.

Level 2

Two booths in your table had different totals but the same unit rate. How is that possible, and what does it tell you about the deals?

- They can have the same unit rate because ____.
- Even though the totals differ, the deals are ____ because ____.

CONNECT

The snack bar sells popcorn as 3 cups for \$2.25 or 8 cups for \$5.60. How do unit rates help you decide the better buy?

Level 1 support

Start here: I find the price per cup for each bag and pick the lower one.

 **Need a hint?**

Sentence starters (optional)

The ____ bag is the better buy because its unit rate is \$ ____ per cup.

La bolsa ____ es la mejor compra porque su tasa unitaria es \$ ____ por taza.

I divided ____ by ____ to find the price per cup.

Dividí ____ entre ____ para hallar el precio por taza.

Word bank: unit rate better buy divide per cost

Listen for: Students compute \$0.75/cup and \$0.70/cup, and choose the large bag because the lower unit rate means each cup costs less.

Level 2

A friend says the large bag must be the better buy because it costs less per cup AND gives more popcorn. Is 'more popcorn' a fair reason to call it the better buy? Justify using the unit rate.

- 'More popcorn' is or is not a fair reason because ____.
- The real reason it is the better buy is ____.

Guided Examples

Example 1

A booth charges \$4.50 for 9 games. What is the unit rate?

Solution: $\$4.50 \div 9 = \0.50 per game.

Answer: A. \$0.50 per game

Example 2

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

Solution: $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon.}$

Answer: A. 30 miles per gallon

Example 3

A pack of 8 markers costs \$6.00. What is the cost per marker?

Solution: $\$6.00 \div 8 = \0.75 per marker.

Answer: A. \$0.75

Write About the Math

The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

1. Kernel Sentence subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.

Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.

Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rate matters in math

Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rate matters in math because ____.

Tasa importa en matemáticas porque ____.

but
pero

Rate matters in math, but ____.

Tasa importa en matemáticas, pero ____.

so
entonces

Rate matters in math, so ____.

Tasa importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question *Pregunta*

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Booth ____ **looks better because** ____.

El puesto ____ *se ve mejor porque* ____.

It is hard to compare because the ____ **are different.**

Es difícil comparar porque los/las ____ *son diferentes.*

The unit rate is ____ **per one** ____.

La tasa unitaria es ____ *por un* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which token pack from the table above is the best deal?

- A. Pack A
- B. Pack B
- C. Pack C
- D. They are all the same

Show your work:

2. Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

- A. Runner B
- B. Runner A
- C. Same rate
- D. Cannot determine

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of \$____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** B. Pack B — *Pack B has a unit rate of \$0.20 per token, which is the lowest cost per token.*
2. **Try It 2:** A. Runner B — *Runner A: $4 \div 12 \approx 0.33$ laps/min. Runner B: $5 \div 14 \approx 0.36$ laps/min. Runner B runs slightly more laps per minute.*
3. **Exit Ticket:** A. 6 for \$1.50 — \$0.25 each — $\$1.50 \div 6 = \0.25 per pencil. $\$2.80 \div 10 = \0.28 per pencil. $\$0.25 < \0.28 , so 6 for \$1.50 is the better buy.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).